Customer Purchase Behavior Analysis Using Covariance and Correlation

Name: Mythrea Shree | Reg no: 192424089 | Course: DSA0405 Fundamentals of data science | Submitted to: Dr Krishnaraj M.

Introduction

Problem
Identify key customer attributes driving purchasing decisions and basket size.

Approach
Apply sample covariance & Pearson correlation matrices to transaction logs.

Why Now?
E-commerce expansion & demand for hyper-personalized marketing systems.

Dataset Architecture (~10,000 Customer Profiles)

- Multi-Channel Retail Log: 12 months of aggregated POS & digital checkout sessions covering ~10,000 active customer records.
- Feature Vectors: Purchase Frequency, Average Spending ($), Annual Income, Customer Age, and Product Category Diversity.
- Analytical Objective: Construct full covariance and correlation matrices to identify core revenue drivers and segment profiles.

Method Pipeline

1. POS Logs
2. Clean & Scale
3. Covariance
4. Pearson r

Validation & Statistical Assessment:
- Significance testing ($p < 0.001$) confirms robust dependencies across all behavioral cohorts.

Results & Empirical Findings

Variable	Freq	Spend	Income	Age	CatDiv
Freq	1.00	0.82	0.35	-0.10	0.55
Spend	0.82	1.00	0.48	-0.05	0.61
Income	0.35	0.48	1.00	0.20	0.30
Age	-0.10	-0.05	0.20	1.00	-0.08
CatDiv	0.55	0.61	0.30	-0.08	1.00

Top Correlated Variable Pairs

- Frequency vs Avg Spend
  r = 0.82 (Strong Positive)
- Spend vs Category Diversity
  r = 0.61 (Strong Positive)
- Frequency vs Category Diversity
  r = 0.55 (Moderate Positive)
- Income vs Avg Spend
  r = 0.48 (Moderate Positive)
- Income vs Frequency
  r = 0.35 (Mild Positive)

Pearson Correlation Formula:

$$r = \Sigma(X - \bar{X})(Y - \bar{Y}) \, / \, [\, \sqrt{\Sigma(X - \bar{X})^2} \cdot \sqrt{\Sigma(Y - \bar{Y})^2} \,]$$

R^2 = 0.71 | 71% OF TOTAL SPEND VARIANCE EXPLAINED

- Bivariate linear models confirm that customer purchase frequency combined with product category diversity accounts for 71% of variance in overall order volume and transaction value across cohorts.

Conclusion

1. Purchase frequency and average spend display strong positive correlation (r = 0.82), showing repeat buyers drive store profitability.
2. Income and category diversity co-vary positively with spend, highlighting immediate high-margin cross-sell opportunities.
3. Age shows negligible linear correlation (r = -0.05), indicating behavior metrics drastically outperform static demographics.

Future Directions

- Causal Inference: Incorporate structural causal models and DAGs to establish true causation behind basket sizes.
- Real-Time Streams: Deploy streaming covariance pipelines on live clickstreams for automated dynamic segmentation.
- Predictive AI: Combine correlation weights with gradient-boosted trees (XGBoost) for personalized customer lifetime value predictions.